



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 8 • STANDARD 6.SP.2

Shape of Data Distributions

Lesson 8-7 • Teacher Copy

Name: Type your name

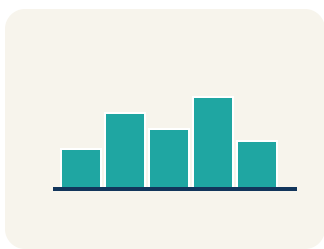
Date: Today's date

Class: Class period

Key Vocabulary

Level 2 Standard

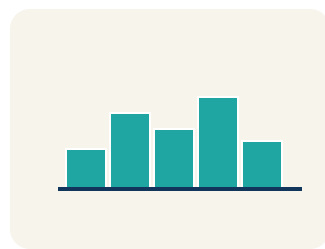
Picture first, then the word, then a plain-language meaning. Say each word out loud.



Heights: 60, 62, 64, 66, 68, 66, 64, 62, 60 — rises and falls evenly like a hill

Symmetric

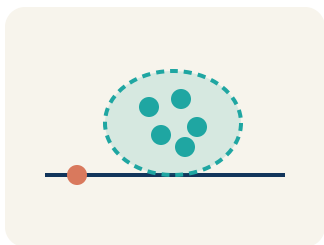
Write the definition:



Skewed right: most data low (1,2,2,3,3,3,15) — tail stretches toward the high value

Skewed

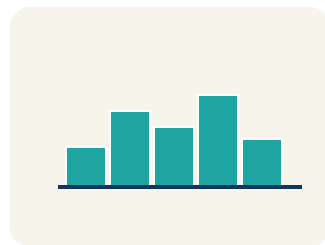
Write the definition:



Scores 18, 19, 20, 21, 22 form a cluster around 20 — they are tightly grouped

Cluster

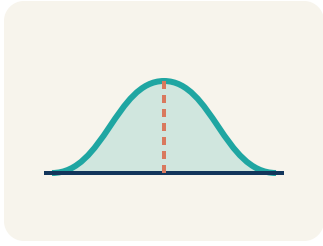
Write the definition:



Data: 5, 6, 7, 8, ____, ____, ____, 20 — the empty space from 9 to 19 is a gap

Gap

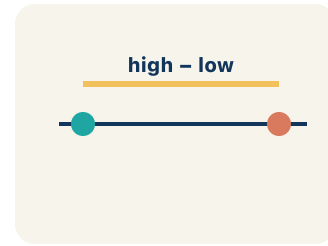
Write the definition:



Looking at a dot plot or histogram, you can see if data is symmetric, skewed, clustered, or has gaps

Data distribution

Write the definition:



Symmetric data with low variability: tightly packed around the center. Skewed data: spread unevenly

Variability

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can describe the shape of a data distribution as symmetric, skewed, or having clusters and gaps.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Symmetric

Skewed

Cluster

Gap

Data distribution

Variability



Tap any word to see what it means and a picture.

1

A distribution where both sides look like mirror images is

2

A distribution with a tail stretching out to one side is .

3

A group of data values bunched close together is a .

4

An interval where there are no data values is a .

5

The way data values are spread out across a range is the .

6

How much the values in a data set differ from one another is the

.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: A histogram of basketball players' ages shows: most players are 22–28, with a long tail of older players up to age 40. What is the shape of this distribution?

- A. Skewed right
- B. Symmetric
- C. Skewed left
- D. Uniform

Step 1 Most data is on the left (younger ages) with a tail stretching right (older ages).

Step 2 This is skewed right.

 **Answer:** A. Skewed right

 Try — put the steps in order

Drag the cards (or use the ▲ ▼ buttons) to put the solution steps in the right order, then press **Check**.

This is skewed right.

Most data is on the left (younger ages) with a tail stretching right (older ages).

 We do — together

Solve this one with your class using the same steps.

Problem: A data set is symmetric. What is likely true about the mean and median?

- A. They are approximately equal
- B. The mean is much higher than the median
- C. The median is much higher than the mean
- D. They are always exactly the same

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Match each description with the correct distribution shape.

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Display 2: A dot plot of starting players' heights has one clear peak in the middle, with roughly even sides falling away from that peak. What shape best describes this distribution?

- A. Symmetric with one cluster around the peak
- B. Skewed right
- C. Skewed left
- D. Has a large gap

Show your work:

2. Display 5: A free-throw percentage histogram has most players clustered between 70% and 85%, a clear empty interval from 50% to 65%, and a few players down at 40%-50%. Which features are present, and what is the overall shape?

- A. A cluster (70-85%), a gap (50-65%), and a left tail — so skewed left
- B. A symmetric display with no gaps
- C. A right tail of high values — so skewed right
- D. Two equal peaks with no cluster or gap

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A teacher shows two dot plots of quiz scores from two different classes. Class A's data is symmetric with a cluster around 80. Class B's data is skewed right with most scores between 60–70 and a few scores near 100. Compare the classes: which has a higher median? Which has a higher mean? In which class does the mean better represent a typical score?

Sentence starter: Class A is ____ with scores clustered around _____. Class B is skewed ____ with most scores at _____. Class A likely has a ____ median. Class B's mean is pulled ____ by the high scores. The mean better represents a typical score in Class ____ because _____.

Show your work:

Reflect — Exit Ticket

A data set has most values clustered between 40–60, with a few values at 90–100. What is the shape of the distribution and which measure of center is best?

- A. Skewed right; use median
- B. Symmetric; use mean
- C. Skewed left; use median
- D. Uniform; use mean

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Symmetric
2. **Notes 2:** Skewed
3. **Notes 3:** Cluster
4. **Notes 4:** Gap
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **We Try (notes):** A. They are approximately equal — *In a symmetric distribution, the mean and median are approximately equal because the data is balanced on both sides.*
8. **Try It 1:** A. Symmetric with one cluster around the peak — *One peak in the middle with roughly even sides means the data mirrors around the center, so it is symmetric with a single cluster at the peak.*
9. **Try It 2:** A. A cluster (70-85%), a gap (50-65%), and a left tail — so skewed left — *The data clusters at the high end (70-85%), there is an empty gap (50-65%), and the few low values form a tail on the left, so the display is skewed left.*
10. **Exit Ticket:** A. Skewed right; use median — *The data clusters on the left (40–60) with a tail to the right (90–100). This is skewed right. The median is best because the mean would be pulled toward the high values.*